TryHackMe Mini CTF Write-Up

Network Enumeration and Service Exploitation

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1 Introduction

This write-up documents the resolution of a mini Capture The Flag (CTF) challenge on TryHackMe. The objective was to enumerate network services, retrieve three hidden keys, and use them to gain access to an administrative environment in order to obtain the final flag.

The challenge focuses on practical usage of common tools such as `nmap`, `netcat`, `dig`, and basic database interaction.

2 Target Information

- **Target IP Address:** `10.64.176.192`
- **Goal:** Retrieve three keys and access the final flag

3 Enumeration

3.1 TCP Port Scanning

The first step consisted of identifying open TCP ports on the target machine.

```
nmap 10.64.176.192
```

By default, Nmap scans the top 1000 TCP ports. The scan revealed two open ports:

- Port 22 (SSH)
- Port 80 (HTTP)

To perform a full TCP scan and gather more information, the following command was used:

```
nmap 10.64.176.192
```

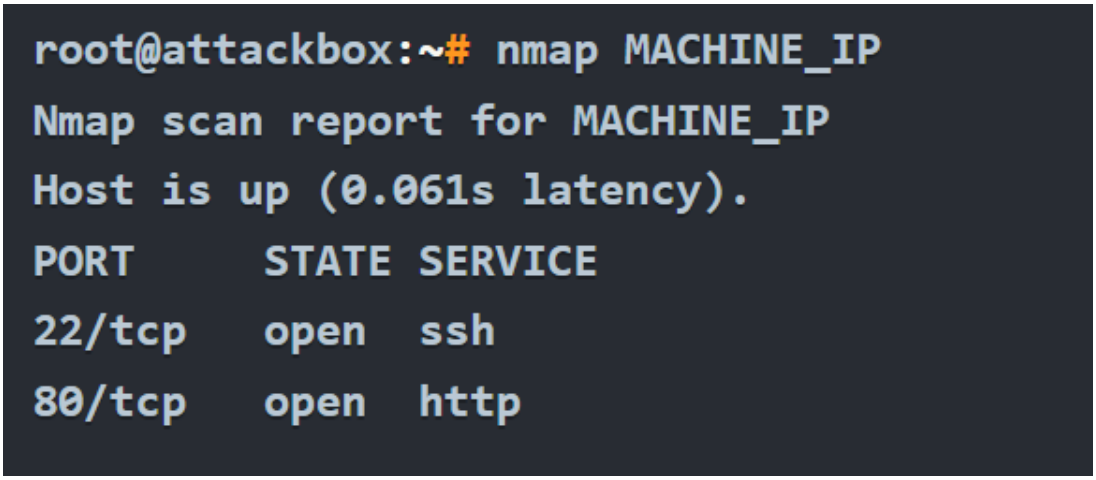

Figure 1: Default Nmap TCP scan (top 1000 ports)

This scan revealed an unusual FTP service running on port **21212** instead of the default port 21.

4 Key 1 – FTP Service

The FTP service discovered on port 21212 was used to retrieve the first key.

```
ftp 10.64.176.192 21212
```

```
root@attackbox:~# ftp MACHINE_IP 21212
Connected to 10.10.239.62.
Name (10.10.239.62:root): anonymous
ftp> ls
-rw-r--r--  1  ftp  ftp  13 Oct 22 16:27 tbfc_qa_key1
ftp> get tbfc_qa_key1 -
[GET HIDDEN KEY 1 FROM HERE]
ftp> ! # Will exit the FTP shell
```

Figure 2: FTP connection and retrieval of the first key

```
ftp 10.64.176.192 21212
```

After connecting successfully, the `GET` command was used to download a file containing the first key.

```
nmap -p- --script=banner 10.64.176.192
```

```
root@attackbox:~# nmap -p- --script=banner MACHINE_IP
Nmap scan report for MACHINE_IP
Host is up (0.00036s latency).
Not shown: 65531 filtered ports
PORT      STATE SERVICE
22/tcp    open  ssh
|_banner: SSH-2.0-OpenSSH_9.6p1 Ubuntu-3ubuntu13.14
80/tcp    open  http
21212/tcp open  trinket-agent
|_banner: 220 (vsFTPd 3.0.5)
25251/tcp open  unknown
|_banner: TBFC maintd v0.2\x0AType HELP for commands.
```

Figure 3: Full TCP scan revealing non-standard FTP service

5 Key 2 – Custom Network Service

A non-standard service was discovered running on port **25251**. To interact with it, `netcat` was used.

```
nc -v 10.64.176.192 25251
```

The verbose option allowed immediate feedback from the service, which directly returned the second key.

```
nc -v 10.64.176.192 25251
```

```
root@attackbox:~# nc -v MACHINE_IP 25251
Connection to 10.10.239.62 25251 port [tcp/*] succeeded!
TBFC maintd v0.2
Type HELP for commands.
HELP                                  # Your input/request
Commands: HELP, STATUS, GET KEY, QUIT  # Server response
[RUN THE CORRECT COMMAND TO GET HIDDEN KEY 2]
```

Figure 4: Netcat interaction returning the second key

6 Key 3 – DNS Enumeration

UDP ports were then scanned to identify additional services.

```
nmap -sU 10.64.176.192
```

```
root@attackbox:~# nmap -sU MACHINE_IP
PORT    STATE SERVICE
53/udp open  domain
[...]
```

Figure 5: UDP scan revealing DNS service on port 53

The scan revealed an open DNS service on port 53.

To query the DNS server and retrieve the third key, the `dig` command was used:

```
dig @10.64.176.192 TXT key3.tbfc.local +short
```

This command requests the TXT record associated with the domain and returns the third key directly.

7 Local Enumeration

After obtaining the three keys, access to the server was achieved. Local services were enumerated using the following command:

```
ss -tunlp
```

This command lists all listening TCP and UDP services along with their associated processes.

```
tbfcapp@tbfc-devqa01:~$ ss -tunlp
Netid  State   Recv-Q  Send-Q       Local Address:Port      Peer Address:Port  Process
udp    UNCONN  0       0                  0.0.0.0:53             0.0.0.0:*
udp    UNCONN  0       0        10.10.76.132%ens5:68             0.0.0.0:*
tcp    LISTEN  0       50                 0.0.0.0:25251          0.0.0.0:*
tcp    LISTEN  0       32                 0.0.0.0:21212          0.0.0.0:*
tcp    LISTEN  0       4096               0.0.0.0:22             0.0.0.0:*
tcp    LISTEN  0       32                 0.0.0.0:53             0.0.0.0:*
tcp    LISTEN  0       511                0.0.0.0:80             0.0.0.0:*
tcp    LISTEN  0       2048             127.0.0.1:8000           0.0.0.0:*
tcp    LISTEN  0       151              127.0.0.1:3306           0.0.0.0:*
tcp    LISTEN  0       4096             127.0.0.1:7681           0.0.0.0:*
tcp    LISTEN  0       4096                 [::]:22                [::]:*
```

Figure 6: Local listening services enumeration

Particular attention was given to port **3306**, which corresponds to the MySQL service.

8 Final Flag – MySQL Access

Using MySQL, the database was queried to retrieve the final flag:

```
mysql -D tbfcqa -e "SELECT * FROM flags;"
```

The query returned the final flag successfully.

```
tbfcapp@tbfc-devqa01:~$ mysql -D tbfcqa01 -e "show tables;"
+--------------------+
| Tables_in_tbfcqa01 |
+--------------------+
| flags              |
+--------------------+
tbfcapp@tbfc-devqa01:~$ mysql -D tbfcqa01 -e "select * from flags;"
[GET YOUR FLAG HERE]
```

Figure 7: Final flag retrieved from MySQL database

9 Conclusion

This mini CTF demonstrated the importance of thorough network enumeration and the ability to interact with both standard and non-standard services. By combining TCP, UDP, and local enumeration techniques, all keys and the final flag were successfully retrieved.